



MATHEMATICS

0580/43

Paper 4 (Extended)

October/November 2019

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **8** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	1254	2	M1 for $342 \div 3$
1(a)(ii)	27.3 or 27.27...	1	
1(b)	867	2	M1 for $1020 \times \frac{15}{100}$ oe or $1020 \times \left(1 - \frac{15}{100}\right)$ oe
1(c)	4.5[0]	3	M2 for $\frac{79.5[0]}{100+6} [\times 6]$ oe or $\frac{79.5[0]}{100+6} \times 100$ oe or M1 for 79.5[0] associated with 106[%]
1(d)	22.6 or 22.58... nfw	4	M1 for $\frac{45}{20}$ or better and M2 for $\frac{60+45}{\text{their } 2\text{h } 24\text{min} + \text{their } \frac{45}{20}}$ or M1 for $\text{their } \frac{45}{20} + \text{their } 2\text{h } 24\text{min}$
1(e)	91.6[0] to 91.61	3	M2 for $480 \times \left(1 + \frac{2.1}{100}\right)^4 - 430$ oe OR M1 for $480 \times \left(1 + \frac{2.1}{100}\right)^4$ oe A1 for 522, 521.6[0] to 521.61
1(f)	112.8125	2	B1 for 2.5 or 9.5 seen

Question	Answer	Marks	Partial Marks
2(a)(i)	$2a + a + 2b + 3b + 10 = 180$ leading to $3a + 5b = 170$ without error or omission	1	
2(a)(ii)	$8a + 3a + 2b + b + 50 + 4b - 2a = 360$ leading to $9a + 7b = 310$ without error or omission	1	
2(a)(iii)	Correct method to eliminate one variable	M1	
	$[a =]15$ $[b =]25$	A2	A1 for each correct value If 0 scored, SC1 for two values that satisfy one of the equations or for two correct answers with no/incorrect working
2(a)(iv)	30	1	
2(b)	-1.5 or $-1\frac{1}{2}$ or $-\frac{3}{2}$	2	M1 for $6x = -12 + 3$ or better
2(c)	$\frac{3x+3}{2}$ oe final answer	3	M1 for $8x - 2y = 5x - 3$ or $4x - y = \frac{1}{2}(5x - 3)$ M1FT for isolating the y term correctly
2(d)	$9x^6$	2	M1 for $(3x^3)^2$ or $(729x^{18})^{\frac{1}{3}}$ seen or for $9x^k$ or kx^6 as final answer
2(e)	$\frac{x}{x-5}$ final answer nfw	3	M1 for $x(x+5)$ M1 for $(x-5)(x+5)$
3(a)	5, -3, 21	3	B1 for each
3(b)	Fully correct curve	4	B3 FT for 9 or 10 points or B2 FT for 7 or 8 points or B1 FT for 5 or 6 points
3(c)	-2.9 to -2.7 0 1.7 to 1.9	2	B1 for 2 correct values

Question	Answer	Marks	Partial Marks
3(d)	Tangent ruled at $x = 2$	B1	
	10 to 14	B2	Dep on correct tangent or close attempt at tangent at $x = 2$ M1 for rise/run also dep on correct tangent drawn or close attempt at tangent Must see correct or implied calculation from a drawn tangent
3(e)	6	1	
4(a)	36.8 or 36.84...	2	M1 for $\frac{h}{107} = \tan 19$ or $\frac{h}{\sin 19} = \frac{107}{\sin 71}$ oe or better
4(b)	42.1 or 42.12... from cosine rule	4	M2 for $[\cos BAC =] \frac{158^2 + 132^2 - 107^2}{2 \times 158 \times 132}$ or M1 for implicit version A1 for $[\cos BAC =] \frac{30939}{41712}$ or 0.7417...
4(c)	35.8 or 35.84... from sine rule	3	M2 for $\frac{86 \times \sin 116}{132} [= 0.58557...]$ or M1 for $\frac{\sin CAD}{86} = \frac{\sin 116}{132}$ oe
4(d)	9670 or 9669 to 9676	3	M2 for $\frac{1}{2} \times 158 \times 132 \times \sin(\text{their (b)})$ oe and $\frac{1}{2} \times 86 \times 132 \times \sin(64 - \text{their (c)})$ oe or M1 for either area
4(e)	214.2 or 214.1... or 214	2	M1 for $[180 +]70 - \text{their (c)}$ oe
5(a)(i)	52	1	
5(a)(ii)	36	1	
5(a)(iii)	26	1	FT 62 – <i>their</i> (a)(ii) evaluated correctly
5(b)	Valid comment	1	Strict FT <i>their</i> (a)(iii), e.g. distances for females are more varied
5(c)	$\frac{11}{20}$ oe	2	M1 for 27 written or answer of $\frac{27}{60}$ oe
5(d)(i)	[18 9] 14 12 5 [2]	2	B1 for 1 correct value

Question	Answer	Marks	Partial Marks
5(d)(ii)	48.75 nfw	4	M1 for midpoints soi M1 for use of $\sum fx$ with <i>their</i> frequencies M1 (dep on 2nd M1) for $\sum fx \div (60$ or by <i>their</i> $\sum f)$
6(a)(i)	Angle $ABC=52$ nfw	B1	ALTERNATIVE [Reflex] angle $AOC = 256$
	Opposite angles in cyclic quad oe Angles in opposite segments	B1	Angle at centre = $2 \times$ angle at circumference/arc
	[Angle $AOC=104$] Angle at centre = $2 \times$ angle at circumference/arc nfw	B1	Angles around a point
6(a)(ii)	22 nfw	2	B1 for angle $OAC = 38$ or angle $CAD = 24$
6(a)(iii)	28	1	
6(a)(iv)	36.6 or 36.62 to 36.63 nfw	3	B2 for 7.4 or 17.42 to 17.43 or M2 for $9.6 \times 2 + \frac{104}{360} \times 2 \times \pi \times 9.6$ or M1 for $\frac{104}{360} \times 2 \times \pi \times 9.6$
6(b)(i)	81	3	M2 for $\frac{A}{36} = \left(\sqrt[3]{\frac{2187}{648}} \right)$ oe or better or for $A \times \frac{648}{36} \times \sqrt[3]{\frac{2187}{648}} = 2187$ oe or better or M1 for $\frac{A^3}{36^3} = \frac{2187^2}{648^2}$ oe or $\sqrt[3]{\frac{2187}{648}}$ or $\sqrt[3]{\frac{648}{2187}}$
6(b)(ii)	8.05 or 8.051 to 8.052...	3	M2 for $\left[r^3 = \right] \frac{2187 \times 3}{4 \times \pi}$ oe or M1 for $\frac{4\pi r^3}{3} = 2187$ SC2 for $\frac{648 \times 3}{4 \times \pi}$ or SC1 for $\frac{4\pi r^3}{3} = 648$
7(a)	Reflection $y = -1$	2	B1 for each
7(b)(i)	Image at $(-6, 5) (-6, 7) (-5, 7) (-4, 5)$	2	B1 for translation by $\begin{pmatrix} -3 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 4 \end{pmatrix}$

Question	Answer	Marks	Partial Marks
7(b)(ii)	Image at (1, -1) (3, -1) (3, -3) (2, -3)	2	B1 for shape correct size and orientation but wrong position
7(b)(iii)	Image at (1, 2) (1, 6) (3, 6) (5, 2)	2	B1 for shape correct size and orientation, wrong position
8(a)(i)	$\frac{2}{5}$ oe	2	M1 for $\frac{4}{6} \times \frac{3}{5}$
8(a)(ii)	$\frac{3}{5}$ oe	1	FT 1 – <i>their</i> $\frac{12}{30}$ oe
8(b)	$\frac{5}{7}$ oe nfwf	4	M3 for $\frac{2}{7} + \frac{5}{7} \times \frac{2}{6} + \frac{5}{7} \times \frac{4}{6} \times \frac{2}{5}$ oe or for $1 - \frac{5}{7} \times \frac{4}{6} \times \frac{3}{5}$ oe or M1 for each of $\frac{5}{7} \times \frac{2}{6}$ and $\frac{5}{7} \times \frac{4}{6} \times \frac{2}{5}$ oe or completed tree diagram with appropriate probabilities shown
9(a)(i)	5	1	
9(a)(ii)	1	2	M1 for $h(0)$ or 3^{9-x^2} or better
9(a)(iii)	$9 - 4x^2$ final answer	1	
9(a)(iv)	$15 - 2x^2$ final answer	2	M1 for $2(9 - x^2) - 3$ or better
9(b)	$\frac{x+3}{2}$ final answer	2	M1 for $x = 2y - 3$ or $y + 3 = 2x$ or better or $\frac{y}{2} = x - \frac{3}{2}$
9(c)	1.8 or $1\frac{4}{5}$ or $\frac{9}{5}$	2	M1 for $10x - 15 = 3$ or $2x - 3 = \frac{3}{5}$
9(d)	-1 and 4 nfwf	4	M1 for $9 - (2x - 3)^2 = -16$ A1 for $4x^2 - 12x - 16 = 0$ oe M1 (dep on first M1) for correct factors or use of formula or completing the square for their 3-term quadratic OR M1 for $9 - y^2 = -16$ A1 for $y^2 = 25$ M1 (dep on first M1) for $2x - 3 = \pm 5$
9(e)	$\frac{1}{9}$	1	

Question	Answer	Marks	Partial Marks
10	$x + 1 - 2x = 3x(x + 1)$	M2	M1 for a common denominator of $x(x + 1)$ seen or attempt to multiply through by denominators or for $\frac{x+1-2x}{x(x+1)} = 3$
	$3x^2 + 4x - 1 [= 0]$ oe nfw	A1	
	$[x =] \frac{-4 \pm \sqrt{4^2 - 4 \times 3 \times (-1)}}{2 \times 3}$	B2	B1FT for $\sqrt{4^2 - 4 \times 3 \times (-1)}$ or better or for $\left(x + \frac{2}{3}\right)^2$ B1FT for $\frac{-4 + \sqrt{q}}{2 \times 3}$ or $\frac{-4 - \sqrt{q}}{2 \times 3}$ or for $-\frac{2}{3} \pm \sqrt{\frac{1}{3} + \left(\frac{2}{3}\right)^2}$
	-1.55 and 0.22 final answers	B2	B1 for each or B1 for -1.548 to -1.549 and 0.215... or for -1.55 and 0.22 seen in working or for -0.22 and 1.55 as final answer or for -1.5 or -1.54 and 0.2 or 0.21 as final answer
11(a)(i)	$8\mathbf{b} - 4\mathbf{a}$ oe	1	
11(a)(ii)	$6\mathbf{b}$	1	
11(a)(iii)	$6\mathbf{b} - 2\mathbf{a}$ or $2(3\mathbf{b} - \mathbf{a})$	1	FT $-2\mathbf{a} + \text{their (a)(ii)}$
11(b)	2 : 1 oe final answer	3	Dep on correct $\overrightarrow{BC}$ or correct $\overrightarrow{AC}$ seen B2 for $\overrightarrow{BC} = 4\mathbf{b} - 2\mathbf{a}$ or M1 for a correct route for $\overrightarrow{BC}$ in terms of a and b or for a correct route for $\overrightarrow{AC}$ in terms of a and b If no/incorrect working seen then SC1 for final answer of 2 : 1 (oe)